Exp No: 01
Date : 23/7/26

Basic Networking commands.

1. arp -a

Displays the ARP, cache, showing the IP address and corresponding MAC address of devices on the local network.

O/P:

-gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3

2. hostname

display the name of our computer.

O/P:

localhost.localdomain

3. ipconfig/all

Displays complete TCP/IP network configuration, including IP address, subnet mask, gateway, DNS, DHCP and MAC address.

O/P:

Windows IP configuration

Host name . . . . . : DEEPIKA
Primary Dns Suffix . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled . . . . . : No
WINS Proxy Enabled . . . . : No

4. nbtstat -a

Displays the NetBIOS name table of a remote computer to help troubleshoot NetBIOS name resolution.

O/P:

Displays protocol statistics and current TCP/IP connections using NBT

NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]

-a (adapter status) Lists the remote machine's name table given its name

-A (Adapter Status) List the remote machine's name table given its IP address

-n (names) Lists local NetBIOS names.

5. netstat

Displays active network connections, listening, ports and network statistics.

O/P:

Active Connections

Proto	Local Address	Foreign Address	State
TCP	[::1]:1521	DEEPIKA:49672	ESTABLISHED
TCP	[::1]:49672	DEEPIKA:1521	ESTABLISHED

6. netstat -r

Displays the routing table of the computer

7.

8.

Interface list

21...00 ff e6 da 62 66.... TAP-Windows Adapter V9
6...3e 8d 3d 7f 5d 30... Realtek RTL8851BE WiFi 6
802.11ax PCIe Adapter #3
8...3a 8d 3d 7f 5d 30... Realtek RTL8851BE WiFi 6
802.11ax PCIe Adapter #5
8...38 8d 3d 7f 5d 30... Realtek RTL8851BE WiFi 6
802.11ax PCIe Adapter
1..... Software Loopback
Interface 1.

7. nslookup

finds the IP address of a domain name using the DNS Server.

o/p:

Default Server: Unknown
Address: 127.0.0.1

8. pathping

Traces the route to a destination & measures packet loss at each network hop.

o/p:

Usage: pathping [-g host-list] [-h maximum_hops]
[-i address] [-n] [-p period]
[-q num_queries] [-w timeout]
[-4] [-6] target_name.

9. Ping

Test connectivity b/w your computer & another computer or website using ICMP

O/P:

Ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS] [-r count] [-s count] [[-j host-list], [-k host-list]] [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p] [-4] [-6] target_name.

10. route

Displays the IP routing table used by the operating system to route network traffic

O/P:

ROUTE [-f] [-p] [-4|-6] command [destination] [MASK netmask] [gateway] [METERIC metric] [IF interface]

Linux Networking commands

1. ip

Display IP address
configure network interface
Add/Delete IP address
Manage routing table.

i. Display IP Address

ip

Displays all network interfaces and their assigned IP addresses

O/P:

Usage: ip [OPTIONS] object {COMMAND | help}
ip [-force] -batch filename.

1. lo: <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc noqueue state unknown group default qlen 1000

inet 127.0.0.1

2. enp0s3:

inet 10.0.2.15/24.

ii. ip link show

Displays the status of all network interfaces.

O/P:

1. lo: <LOOPBACK, UP, LOWER_UP>

2. enp0s3: <BROADCAST, MULTICAST, LOWER_UP>

iii. ip route

Displays the routing table & default gateway

O/P:

10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src

10.0.2.15 metric 100

2. ip address add

Assigns a new IP address to a network interface.

Ex: ip address add 192.168.1.254/24

O/P:

1. lo: <LOOPBACK, UP, LOWER_UP>
2. enp0s3: <BROADCAST, MULTICAST, UP, LOWER_UP>
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3.
   inet 192.168.1.254/24 scope global enp0s3.

3. Delete an IP Address

Remove the IP address from the interface

Eg: ip address del 192.168.1.254/24 dev enp0s3.

O/P:

1. lo: <LOOPBACK, UP, LOWER_UP>
2. enp0s3: <BROADCAST, MULTICAST, UP, LOWER_UP>
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3.

1. Bring Interface Online

Enables the Network interface.

Eg: ip link set enp0s3 up

O/P:

1. lo: mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT mode group default qlen 1000 link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

2. enp0s3: mtu 1500 qdisc fq_codel state
up mode Default group default qlen 1000 link/ether
08:00:27:55:b8:a1 brd ff:ff:ff:ff:ff:ff

5. Interface Offline

Displays the network interfaces.

Eg: ip link set enp0s3 down

O/P: 1. lo: <loopBACK, UP, LOWER_UP>

2. enp0s3: <BROADCAST, MULTICAST, DOWN, LOWER_UP>,

6. Enable Promiscuous Mode.

Allows the interface to receive all network packets

Eg: ip link set enp0s3 promisc on

O/P: 1. lo : <loopBACK, UP, LOWER_UP>

2. enp0s3: <BROADCAST, MULTICAST, PROMISC>

7. Add a default route

Adds the default gateway for all network traffic.

Eg: ip route add default via 192.168.1.254 dev enp0s3

O/P:

default via 198.168.1.254 dev enp0s3

192.168.10/24 dev enp0s3 proto kernel scope link src 192.168.1.15.

8. Add a Route through a Gateway.

Route traffic for the 192.168.1.0/24 network through gateway 192.168.1.254

O/P:

default via 192.168.1.254 dev eth0

192.168.1.0/24 via 192.168.1.254 dev eth0

192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.100

9. Add a route via Interface.

Add route to the 192.168.10/24 network through the interface eth0.

Eg: ip route add 192.168.1.0/24 dev eth0

O/P:

10. Delete a Route

Deletes the route to the 192.168.1.0/24 network through gateway 192.168.1.0/24.

Eg: ip route delete 192.168.1.0/24 via 192.168.1.254

11. ifconfig

It is a staple in many sysadmin's tool belt for configuration configuring & Troubleshooting Networks.

O/P:

Windows IP configuration

Unknown adapter McAfee VPN:

Media State . . . . . : Media disconnected
Connection-Specific DNS Suffix . :

Wireless LAN adapter Wi-Fi 3:

Media State . . . . . : Media disconnected
Connection-Specific DNS Suffix . :

18. mtr

combines the functionality of the ping and Traceroute commands.

a) mtr google.com

Shows you the statistics, including each hop with time & loss %.

o/p:

Host	Loss%	Snt	Last	Avg	Best
1. 172.16.8.1	0.0%	134	0.1	0.2	0.1
2. 172.91.163.199	8.6%	134	3.9	5.3	1.1
3. ???	2.2%	188	46.6	18.6	4.1

b) mtr -g google.com

Show numeric IP address

c) mtr -b google.com

Show numeric IP address & hostnames.

d) mtr -c 10 google.com

Set the number of ping that you want.

17. tcpdump

This command is designed for capturing & displaying packets.

i) tcpdump -i eth0 -c 10 host 8.8.8.8

Filter our traffic comming from a specific host.

18. ping DEEPIKA.

It display best way to ~~display~~ test connectivity b/w two nodes.

Syntax: ping hostname

O/P:

Pinging DEEPIKA [fe80::591f:7ba0:f461:c63%.2] with 32 bytes of data:

Reply from fe80::591f:7ba0:f461:c63%.2: time<1

Ping statistics for fe80::591f::7ba0:f461:c63%.2

Packets: Sent = 4, Received = 4, Lost = 0 (0% Loss),

Approximate round trip time in mili-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms.

Result:

The Basic Networking & Linux commands are Learned & executed Successfully.